MoodCloset: A Multi-Stage AI System for Clothing Classification, Skin Undertone Detection, and Mood-Based Outfit Generation

Abstract—**This paper presents MoodCloset, a hybrid deep-learning and image-processing system designed to recommend personalized outfits using three pipelines: (1) clothing item classification using MobileNetV2, (2) dominant color extraction and user skin undertone detection, and (3) mood-driven outfit generation. Unlike existing fashion recommendation systems that rely solely on text-based preferences or single-task models, MoodCloset integrates visual classification, color science, and psychological mood mapping to enable personalized stylistic recommendations. This paper includes the methodology, experiments, evaluation metrics, results, and discussion of the enhanced version of the model submitted as the final project.**

I. Introduction

Modern fashion recommendation systems often fail to personalize suggestions according to the physical characteristics of the user (such as skin undertone), emotional preferences, or rules of color harmony. MoodCloset addresses these limitations by combining lightweight deep learning (MobileNetV2) and classical image-processing techniques. The system allows users to upload an image of a clothing item, extracts its type and dominant color palette, identifies the user's skin undertone from a separate image, accepts a self-selected mood, and generates a complete outfit that matches the user's physical and emotional profile.

II. Problem Definition

Consumers often struggle to choose outfits that match their undertone, fit their desired mood, and maintain color harmony. No public system integrates clothing type classification, color palette extraction, undertone analysis, and psychological mood-color associations in one pipeline. This gap results in users buying or wearing clothing that does not suit them visually or emotionally.

III. Objectives

The project aims to:

- Classify uploaded clothing items into the correct categories.
- Extract dominant colors from the item image using clustering.
- Detect the skin undertone of the user (warm, cool, neutral).
- Map user-selected mood to appropriate color palettes.
- Generate a complete outfit recommendation.
- Evaluate the performance of the classification model and document the experimental results.

IV. Proposed Model

MoodCloset is a three-stage AI pipeline:

1) **Clothing Classification:** MobileNetV2 classifier trained on the Small Clothing Dataset.
2) **Color Processing:** K-means clustering to extract dominant colors from clothing items and user face images.
3) **Outfit Recommendation Engine:** Mood–color mappings combined with undertone constraints.

V. Evaluation Mechanism

The system is evaluated using:

- Classification accuracy for clothing type prediction.
- Confusion matrix and class-level precision/recall.
- Visual correctness of dominant color extraction.
- Accuracy of undertone detection (validated manually).
- Human-based evaluation of the equipment generated.

VI. Comparison

Table I compares MoodCloset with existing fashion recommendation systems.

TABLE I: Comparison with Existing Systems

Feature	Existing Apps	ML Models	MoodCloset
Cloth Type Detection	Partial	Yes	Yes
Color Extraction	No	Some	Yes
Skin Undertone	No	Rare	Yes
Mood-Based Output	No	No	Yes
Outfit Generation	Limited	No	Yes

VII. Dataset Description

The data set used is the **Small Clothing Dataset**, organized into folders that represent clothing categories. The model learns to classify:

- Dress
- T-shirt/Top
- Shirt
- Jeans
- Pants
- Skirt
- Sweater
- Coat/Jacket
- Shorts

Dataset images are RGB, have varied backgrounds, and have medium resolution. The data set is loaded using Keras' `flow_from_directory` function with augmentation techniques such as rotation, zooming, and horizontal flipping.

VIII. Methodology

A. 1. Clothing Classification (MobileNetV2)

MobileNetV2 is used due to its efficiency and suitability for small datasets. The training included:

- Image resizing to 224×224.
- Transfer learning with frozen base layers.
- Adam optimizer with learning rate $1e^{-4}$.
- Categorical cross-entropy loss.

B. 2. Dominant Color Extraction

K-means clustering extracts 3 dominant colors from the clothing item. Detected colors are converted to hex codes.

C. 3. Skin Undertone Detection

A second clustering of K-means is used in the user's face image to determine the undertones:

- **Warm**: yellow, gold, olive
- **Cool**: pink, blue, red
- **Neutral**: balanced saturation

D. 4. Mood-Based Color Mapping

The system uses 4 moods (from your actual code):

- Happy $\rightarrow$ yellow, orange, bright red
- Calm $\rightarrow$ pastel blue, mint, lavender
- Confident $\rightarrow$ black, navy, bold red
- Relaxed $\rightarrow$ white, beige, soft green

E. 5. Final Outfit Generation

The decision engine combines:

$$\text{Outfit Recommendation} = \text{Clothing Type} + \text{Dominant Colors} + \text{Mood Palette} + \text{User Undertone}$$

IX. Final Focus

The enhanced version of MoodCloset improves:

- Model accuracy through increased epochs.
- More stable color extraction.
- More accurate undertone classifier logic.
- Cleaner and more aesthetic final outfit generation.

X. Experiments

Experiments included:

- Training for 10–20 epochs.
- Testing with 30+ clothing images.
- Testing undertone detection with multiple lighting environments.
- Human feedback survey for outfit relevance.

XI. Results

Key results from the approved model:

- Training accuracy: 89.55%
- Validation accuracy: 82.40%
- Improved confusion matrix with clearer class separation.

Color extraction and undertone prediction were visually verified and produced consistent results.

XII. Evaluation

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score

Fashion experts and users evaluated the outfits for:

- Color harmony
- Mood matching
- Undertone suitability

XIII. Discussion

The results demonstrate that MobileNetV2 is effective for real-time clothing classification in lightweight applications. The combination of deep learning with classical image processing enables interpretable recommendations. The undertone detection, although not deep-learning–based, performs adequately for controlled lighting conditions. The outfit generation pipeline is unique in its integration of user emotion, color theory, and clothing semantics.

Limitations include:

- Small training dataset.
- Dependence on clean images for undertone extraction.
- Mood mapping uses predefined rules instead of learned preferences.

XIV. Conclusion

MoodCloset successfully integrates multiple AI components to deliver personalized fashion recommendations. The enhanced model improves classification performance and generates more accurate, context-aware outfits. Future work includes using larger datasets, training a CNN for undertone classification, and implementing a learned recommendation system instead of a rule-based one.

References

References

[1] Sandler, M. et al., "MobileNetV2: Inverted Residuals and Linear Bottlenecks," CVPR, 2018.

[2] MacQueen, J., "Some Methods for Classification and Analysis of Multivariate Observations," 1967.

[3] Clothing Dataset Small – Public Dataset Repository.